

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-22304666})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary ideal I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{(1-\sigma_x)^2}(t) J \mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}1025 + \mathbf{Z}(922 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K . The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 22304666 = 0$, of discriminant -89218664 . Class group invariants $(310 \ 5 \ 5)$, class number 7750; the three stored generator ideals (HNF) are $\begin{pmatrix} 237 & 164 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 575)y^3 + (24s + 97717)y^2 + (-43s + 1659874)y + (-1614s + 1807464)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 1149y^8 + 196584y^7 + 25759605y^6 + 958544598y^5 \\ & + 18565428401y^4 + 204280939244y^3 + 1421675858334y^2 + 909630185 \\ & 8536y + 61370491822632 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right)s, \quad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2011971826836 = 2^2 \cdot 3^8 \cdot 19 \cdot 59 \cdot 68389$:

$$\begin{pmatrix} 4139859726 & 343395000 & 2535244326 & 2063698974 & 382998456 & 3977215220 & 2673539728 & 3540337911 & 4022145501 & 2141703746 \\ 0 & 3 & 0 & 0 & 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 9 & 6 & 6 & 2 & 3 & 8 & 1 & 5 \\ 0 & 0 & 0 & 3 & 0 & 0 & 2 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 6 & 3 & 1 & 1 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 165 factors with signed exponents of absolute value up to 4641384731; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 575)y^3 + (24s + 97717)y^2 + (-43s + 1659874)y + (-1614s + 1807464)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 1149y^8 + 196584y^7 + 25759605y^6 + 958544598y^5 \\ & + 18565428401y^4 + 204280939244y^3 + 1421675858334y^2 + 909630185 \\ & 8536y + 61370491822632 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right)s, \quad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3086462423376 = 2^4 \cdot 3^3 \cdot 19 \cdot 199 \cdot 1889603$:

$$\begin{pmatrix} 128602600974 & 96613396386 & 43447505518 & 91124890311 & 5974728758 & 101959981973 & 110677499833 & 68379935548 & 48100140050 & 124877916841 \\ 0 & 6 & 2 & 5 & 2 & 5 & 3 & 1 & 3 & 5 \\ 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 877492637; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 575)y^3 + (24s + 97717)y^2 + (-43s + 1659874)y + (-1614s + 1807464)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 1149y^8 + 196584y^7 + 25759605y^6 + 958544598y^5 \\ & + 18565428401y^4 + 204280939244y^3 + 1421675858334y^2 + 909630185 \\ & 8536y + 61370491822632 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625} \right) s, \quad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $157110630918021 = 3^6 \cdot 11 \cdot 83 \cdot 236051773$:

$$\begin{pmatrix} 1939637418741 & 1259274219294 & 671565227709 & 1124033623668 & 313731578856 & 1101481365564 & 235373477572 & 791704982717 & 690060676963 & 1931537858284 \\ 0 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 & 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 4176178902; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1283)y^3 + (-49s + 56092)y^2 + (-756s + 3363456)y + (-1548s + 34778840)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 2562y^8 + 117316y^7 + 30453299y^6 + 2098029052y^5 \\ + 81654727066y^4 + 2009646785008y^3 + 31346092288336y^2 + 286159 \\ 962069696y + 1263016472100064$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ -1 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right)s, \quad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $37321958351972 = 2^2 \cdot 7^4 \cdot 163 \cdot 23841011$:

$$\begin{pmatrix} 380836309714 & 349127883664 & 326166156547 & 300503090777 & 260035030516 & 352731983919 & 54382164384 & 221182113499 & 87349334833 & 286488993511 \\ 0 & 14 & 7 & 0 & 6 & 7 & 13 & 4 & 8 & 7 \\ 0 & 0 & 7 & 1 & 1 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 241 factors with signed exponents of absolute value up to 27682657; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1283)y^3 + (-49s + 56092)y^2 + (-756s + 3363456)y + (-1548s + 34778840)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 2562y^8 + 117316y^7 + 30453299y^6 + 2098029052y^5 \\ + 81654727066y^4 + 2009646785008y^3 + 31346092288336y^2 + 286159 \\ 962069696y + 1263016472100064$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ -1 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right)s, \quad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $689321696 = 2^5 \cdot 7 \cdot 113^2 \cdot 241$:

$$\begin{pmatrix} 762524 & 28928 & 743978 & 208950 & 199895 & 283 & 374462 & 522642 & 118486 & 277796 \\ 0 & 226 & 112 & 114 & 113 & 198 & 44 & 0 & 141 & 207 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 237 factors with signed exponents of absolute value up to 23507574; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (4 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1283)y^3 + (-49s + 56092)y^2 + (-756s + 3363456)y + (-1548s + 34778840)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2562y^8 + 117316y^7 + 30453299y^6 + 2098029052y^5 \\ & + 81654727066y^4 + 2009646785008y^3 + 31346092288336y^2 + 286159 \\ & 962069696y + 1263016472100064 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ -1 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right) s, \quad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $29923882795576 = 2^3 \cdot 7^2 \cdot 11 \cdot 6939675973$:

$$\begin{pmatrix} 14961941397788 & 2721761786549 & 4331885870012 & 4996643103853 & 6154131505386 & 4291710260372 & 5299884086854 & 12874011541561 & 12150013129842 & 6491287023458 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 234 factors with signed exponents of absolute value up to 39777566; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 3276y^3 + (32s + 163370)y^2 + (-1628s - 5034956)y + (21360s + 73909920)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 6551y^8 + 333292y^7 + 335524y^6 - 912510488y^5 + \\ & 82370946756y^4 - 4453347079152y^3 + 139107409683120y^2 - 22955148 \\ & 70041600y + 15639171215040000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right) s, \quad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $17333214462000 = 2^4 \cdot 3^4 \cdot 5^3 \cdot 73 \cdot 79 \cdot 18553$:

$$\begin{pmatrix} 96295635900 & 61876954230 & 76593294616 & 46930014924 & 96004604488 & 54109049711 & 39315928898 & 76760808089 & 96249509450 & 92463269999 \\ 0 & 15 & 1 & 12 & 2 & 7 & 7 & 7 & 4 & 10 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 1299592122154; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 3276y^3 + (32s + 163370)y^2 + (-1628s - 5034956)y + (21360s + 73909920)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 6551y^8 + 333292y^7 + 335524y^6 - 912510488y^5 + \\ & 82370946756y^4 - 4453347079152y^3 + 139107409683120y^2 - 22955148 \\ & 70041600y + 15639171215040000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right)s, \quad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25751690626500 = 2^2 \cdot 3^6 \cdot 5^3 \cdot 89 \cdot 793813$:

$$\begin{pmatrix} 19075326390 & 705655890 & 3266055590 & 8336355333 & 8347812428 & 18750583840 & 18771048831 & 14287340942 & 4065395727 & 5743682280 \\ 0 & 45 & 0 & 35 & 44 & 24 & 7 & 10 & 21 & 24 \\ 0 & 0 & 10 & 7 & 2 & 8 & 5 & 6 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 2777918197332; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 3276y^3 + (32s + 163370)y^2 + (-1628s - 5034956)y + (21360s + 73909920)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 6551y^8 + 333292y^7 + 335524y^6 - 912510488y^5 + \\ & 82370946756y^4 - 4453347079152y^3 + 139107409683120y^2 - 22955148 \\ & 70041600y + 15639171215040000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625} \right) s, \quad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $237047164054650 = 2 \cdot 3^2 \cdot 5^2 \cdot 67 \cdot 79 \cdot 7547 \cdot 13187$:

$$\begin{pmatrix} 47409432810930 & 18226675286445 & 19396256919290 & 26564155288417 & 5719458227677 & 14868344071665 & 30874904452391 & 20179779199322 & 28277677276695 & 8799753669678 \\ 0 & 5 & 2 & 1 & 3 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 2706310439783; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4220y^3 + 10sy^2 + 4868019y + 20350s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8440y^8 + 27544438y^6 + 43316546960y^4 + 32775608046361y^2 + 9236864045585000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ -1 \ -2 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right)s, \quad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32380840522000 = 2^4 \cdot 5^3 \cdot 349 \cdot 1697 \cdot 27337$:

$$\begin{pmatrix} 809521013050 & 636914682420 & 703215958438 & 398336920708 & 269685070550 & 705101951840 & 239535646061 & 120960869397 & 82083845502 & 661064393056 \\ 0 & 10 & 6 & 4 & 8 & 8 & 7 & 8 & 2 & 6 \\ 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 200 factors with signed exponents of absolute value up to 10369746757; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4220y^3 + 10sy^2 + 4868019y + 20350s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8440y^8 + 27544438y^6 + 43316546960y^4 + 32775608046361y^2 + 9236864045585000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ -1 \ -2 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right)s, \quad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $105422785037020 = 2^2 \cdot 5 \cdot 11 \cdot 37^2 \cdot 79 \cdot 367 \cdot 12073$:

$$\begin{pmatrix} 1424632230230 & 628032865510 & 422113434189 & 196354615504 & 349002126102 & 1346453271136 & 1004849654733 & 263059059484 & 1177517468987 & 312535138748 \\ 0 & 74 & 1 & 73 & 0 & 46 & 53 & 0 & 0 & 68 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 196 factors with signed exponents of absolute value up to 4307192759; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4220y^3 + 10sy^2 + 4868019y + 20350s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8440y^8 + 27544438y^6 + 43316546960y^4 + 32775608046361y^2 + 9236864045585000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ -1 \ -2 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625} \right) s, \quad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $381290360000 = 2^6 \cdot 5^4 \cdot 11^2 \cdot 78779$:

$$\begin{pmatrix} 86656900 & 46118270 & 51138190 & 67208238 & 6309924 & 23466316 & 78826356 & 55931577 & 10144121 & 4154391 \\ 0 & 110 & 50 & 24 & 84 & 92 & 42 & 86 & 61 & 31 \\ 0 & 0 & 10 & 8 & 8 & 9 & 9 & 3 & 4 & 3 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 202 factors with signed exponents of absolute value up to 181731065; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1052y^3 + 182212y - 576s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2104y^8 + 1471128y^6 - 383374048y^4 + 33201212944y^2 + 7400152866816$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right)s, \quad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2008703366028 = 2^2 \cdot 3^5 \cdot 97 \cdot 21304817$:

$$\begin{pmatrix} 12399403494 & 4050750762 & 5705174976 & 1999300920 & 9282945174 & 8234810009 & 11543884875 & 8933328743 & 11592209811 & 11541840 \\ 0 & 3 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 6 & 3 & 0 & 5 & 0 & 4 & 4 & 3 \\ 0 & 0 & 0 & 3 & 0 & 2 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 275 factors with signed exponents of absolute value up to 28486885; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1052y^3 + 182212y - 576s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2104y^8 + 1471128y^6 - 383374048y^4 + 33201212944y^2 + 7400152866816$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right)s, \quad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9950729965776 = 2^4 \cdot 3^6 \cdot 97 \cdot 8794997$:

$$\begin{pmatrix} 5118688254 & 2052748482 & 1760408034 & 4938845682 & 1543787364 & 4619095182 & 2937414356 & 2024979838 & 2408132052 & 688575088 \\ 0 & 6 & 0 & 0 & 3 & 0 & 5 & 5 & 3 & 5 \\ 0 & 0 & 3 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 & 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 0 & 3 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 264 factors with signed exponents of absolute value up to 72158913; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 4 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1052y^3 + 182212y - 576s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2104y^8 + 1471128y^6 - 383374048y^4 + 33201212944y^2 + 7400152866816$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625}\right)s, \quad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $474945511098312 = 2^3 \cdot 3^2 \cdot 6596465431921$:

$$\begin{pmatrix} 39578792591526 & 779469330886 & 19504140858906 & 9498948272704 & 13057321657732 & 37059068354727 & 10298056479453 & 6709420054922 & 2214858801416 & 34003988981637 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 268 factors with signed exponents of absolute value up to 167281467; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2352y^3 + 24sy^2 + 1606412y + 32s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4704y^8 + 8744728y^6 + 5290925568y^4 + 2614819480720y^2 + 22839977984$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{28132863}{1131408212890625} + \left(\frac{3904}{1131408212890625}\right)s, \quad J = \begin{pmatrix} 1025 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1025$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6131132579636 = 2^2 \cdot 797 \cdot 33343 \cdot 57679$:

$$\begin{pmatrix} 3065566289818 & 1503637013046 & 755449518008 & 518193045176 & 2732433849554 & 2519667887477 & 2652942676250 & 284925175767 & 116629298813 & 2093749454496 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 547 factors with signed exponents of absolute value up to 2253329443234709431183837; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2352y^3 + 24sy^2 + 1606412y + 32s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4704y^8 + 8744728y^6 + 5290925568y^4 + 2614819480720y^2 + 22839977984$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2753668}{1025251253128125} + \left(\frac{8377}{8202010025025000}\right)s, \quad J = \begin{pmatrix} 2010 & 542 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2010$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12579753651008 = 2^6 \cdot 196558650797$:

$$\begin{pmatrix} 786234603188 & 156664761530 & 34451188536 & 181332555294 & 785572568024 & 202059719132 & 246618447494 & 401012949447 & 178928714585 & 387125885272 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 545 factors with signed exponents of absolute value up to 1300856544693427787457525; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2352y^3 + 24sy^2 + 1606412y + 32s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4704y^8 + 8744728y^6 + 5290925568y^4 + 2614819480720y^2 + 22839977984$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{326541083}{124636871337890625} + \left(\frac{28414}{124636871337890625} \right) s, \quad J = \begin{pmatrix} 2625 & 328 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2625$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $433874004813712 = 2^4 \cdot 31 \cdot 874745977447$:

$$\begin{pmatrix} 108468501203428 & 34716882907440 & 31889153178230 & 72488941472930 & 105641363862040 & 15965878080521 & 62553761915726 & 85085457697953 & 16658379344833 & 8394878856656 \\ 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 547 factors with signed exponents of absolute value up to 2243372457587227807225125; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.